Coursework: Competition

50% of your final grade

Overview

This year's coursework asks you to build a **metamodel** on top of a primary trading signal that we provide for **11 instruments across three asset classes**. The metamodel's job is to predict, for each primary signal, the probability that following it would be profitable under a triple-barrier exit rule.

This is a **group project** completed in teams of **5 students** (26 groups in total). The coursework is marked out of **100**.

Key Information

Deadline	June 4, 2026
Team Size	5 students (26 groups)
Weight	50% of final grade
Marked out of	100 (with +10 bonus for the optional competition track)

The Universe

You are provided with the primary model's daily signals (-1, 0, +1) for the following 11 instruments.

Equity Index Futures

Ticker	Index
ES1S	S&P 500
NQ1S	Nasdaq 100
FESX1S	Euro Stoxx 50

Energy

Ticker	Commodity
CL1S	WTI Crude Oil
HO1S	Heating Oil
RB1S	RBOB Gasoline
NG1S	Natural Gas

Metals

Ticker	Metal
GC1S	Gold
SI1S	Silver
HG1S	Copper
PL1S	Platinum

You are required to cover **at least one full asset class**. Covering more (up to all 11 instruments) is optional.

Task Description

Build a metamodel for each instrument you cover. The metamodel takes the primary signal plus your features and outputs a **probability in [0, 1]** that the bet is worth taking.

The pipeline is:

1. Feature engineering from OHLCV (and anything else you can derive)
2. Labeling via the **triple-barrier method**, as taught in the course
3. Training and comparing several ML models with hyperparameter tuning
4. Feature importance analysis at the cluster level
5. Evaluation on a clean out-of-sample period
6. (Optional) Strategy construction on top of the metamodel probabilities

Marking Scheme

Section	Marks
Feature Engineering	20
Labeling (Triple-Barrier Method)	20
Model Development and Comparison	30
Feature Importance Analysis (Cluster-Level)	10
Model Evaluation	20
Total	**100**
Optional: Strategy Construction (Competition)	+10 bonus

The bonus is capped so the final mark does not exceed 100.

1. Feature Engineering (20 marks) 📊

Build a rich feature set drawing on the techniques covered in the course:

- Technical indicators
- Latent variable models (GMM, HMM)
- Any of the unsupervised learning methods we discussed
- Anything else you can justify

Be as creative as possible. Document what each feature is meant to capture.

2. Labeling: Triple-Barrier Method (20 marks) 🏷️

Apply the triple-barrier method as taught in the course. You must justify your choice of barrier widths and time-limit.

3. Model Development and Comparison (30 marks) 🤖

We expect **at least three models with hyperparameter tuning**, drawn from across the three families:

- **Linear models** (e.g. Logistic Regression with regularization)
- **Tree-based models** (e.g. Random Forest, XGBoost, LightGBM)
- **Neural networks** (e.g. Variable Selection Network or Sequential Neural Networks)

Present a clear comparison: which model wins, on which metric, and why you think so.

4. Feature Importance Analysis: Cluster-Level (10 marks) 🔍

Beyond per-feature importance, compute importance at the **cluster level**:

- Cluster correlated features together
- Apply MDI, MDA, or SHAP at the cluster level
- Discuss which feature groups drive your metamodel

5. Model Evaluation (20 marks) 📈

Evaluate on an out-of-sample period that you carve out cleanly from the training period.

- Classification metrics: precision, recall, F1, AUC
- Confusion matrix and decision-threshold analysis
- **Per-instrument breakdown** (the metamodel may help on some instruments and not others, say so)
- Comparison against a baseline that follows the primary signal blindly

Optional: Strategy Construction, Competition Track (10 bonus marks) 💹

For groups that want to compete: use the metamodel probabilities to build a position-sizing strategy on top of the primary signal, either on a single asset class or on the full 11-instrument universe.

Full constraints (position limits, gross/net exposure, rebalancing rules, target volatility) will be released on Wednesday 20 May.

Backtest metrics to report:

- CAGR
- Annualised volatility
- Sharpe ratio
- Sortino ratio
- Maximum drawdown
- Average holding period
- Turnover

Dataset

Two CSV files are available on **Insendi under Coursework**:

`ohlcv_data.csv`

Daily OHLCV history for all 11 instruments. One row per (instrument, date).

Column	Description
`date`	Trading date (YYYY-MM-DD)
`instrument`	Lowercase ticker (e.g. `cl1s`, `es1s`, `gc1s`)
`open`, `high`, `low`, `close`	Continuous-contract prices
`volume`	Daily volume
`open_interest`	Daily open interest

History starts in 1990 for most instruments. Equity Index futures start later: ES1S in 1997, FESX1S in 1998, NQ1S in 1999.

`primary_signals.csv`

Daily primary model signals from January 2020 onwards. One row per date, one column per instrument.

Column	Description
`date`	Trading date (YYYY-MM-DD)

Column	Description
`es1s`, `nq1s`, ..., `pl1s`	Primary signal in {−1, 0, +1}

The signal convention is:

- `+1`: the primary model wants to go **long** that day
- `-1`: the primary model wants to go **short** that day
- `0`: no position taken by the primary model

Important. The data we release covers up to **30 June 2022**. The final **6 months** of data (July to December 2022) are held out and will be used as a hidden test set to evaluate your final submission.

Evaluation

You will be judged on:

1. Quality and creativity of your feature engineering 💡
2. Rigour of your labeling and validation protocol 🛡️
3. Appropriateness of your model selection and comparison 🧠
4. Critical analysis of your results 🔬
5. Code quality, reproducibility, and documentation 📝

The score is focused entirely on methodology, not on performance. You can score a high mark even if your metamodel does not beat the primary signal.

🌟🌟🌟 The best submission, judged on **quality of research** rather than performance, will be **presented to the research team at Alken Asset Management, with an interview for an internship at the end of it.**

Getting Started

OHLCV Data

Download `ohlcv_data.csv` from the Coursework folder on Insendi.

Primary Signals

Download `primary_signals.csv` from the Coursework folder on Insendi.

Programming Sessions

For implementation guidance, refer to **all programming sessions and optional programming sessions** of the course.

Submission Rules

- **Group size**: 5 students per group, 26 groups in total
- **One submission per group**: a single combined submission
- **Documentation**: code must be clean, well-documented and reproducible, your notebook should run end-to-end and produce the deliverable CSV
- **Academic integrity**: all work must be original. Plagiarism will result in zero marks and potential disciplinary action.

Deliverables

1. Code

A Jupyter notebook or a set of Python files that runs end-to-end and produces the deliverable files below. **Clean, well-documented code is part of the mark**.

2. Required: Metamodel Predictions

A CSV file covering the **first half of 2022** (January to June). We will rerun your code on the hidden second half of 2022 for the final test.

Format: one row per (date, instrument, prediction).

```
date,instrument,prediction
2022-01-03,cl1s,0.74
2022-01-03,es1s,0.51
...
```

- `date`: trading date (YYYY-MM-DD)
- `instrument`: lowercase ticker
- `prediction`: probability in [0, 1] that the primary signal is worth taking

3. Optional: Strategy Weights

For groups competing on the strategy track, an additional CSV covering the first half of 2022:

```
date,instrument,weight
2022-01-03,cl1s,0.18
2022-01-03,es1s,-0.05
...
```

- `weight`: signed position weight (positive = long, negative = short)

Constraints on the weights will be specified on **20 May**.

Contact

For any question regarding the coursework: h.madmoun@ic.ac.uk

Good luck and have fun!